EXPERIMENT 1 — 2D SKETCH OF GIVEN COMPONENT

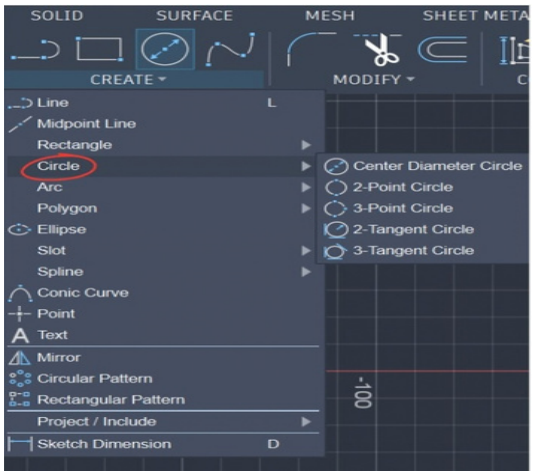
AIM:
To sketch the given drawing with dimensions using fusion 360 tool.

REQUIREMENTS:

1. Autodesk fusion 360
2. Windows 10
3. 16 GB of RAM (integrated graphics recommend 2 GB or more) 4.
2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE:

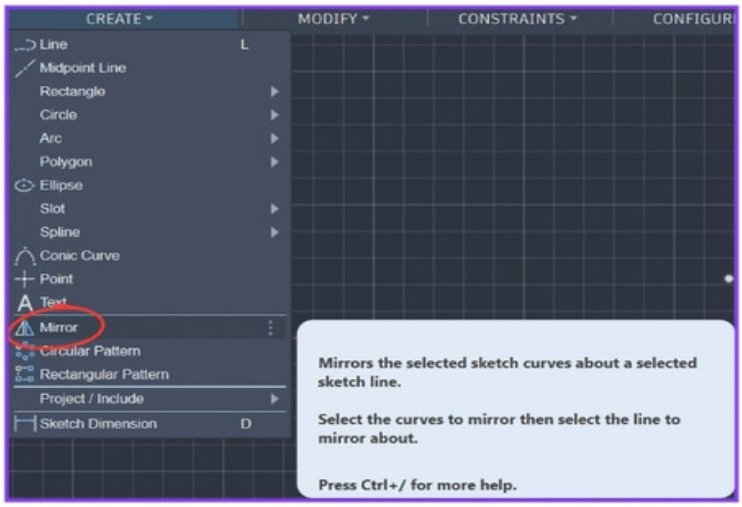
1. Open **Autodesk Fusion** and create a new design. Select **Create Sketch** from the Sketch section and choose the **Front plane** as the sketching plane.
2. After entering the sketch environment, select the **Line** tool from the Sketch toolbar. Draw the initial horizontal construction according to the given drawing and apply the required dimensions using the **Sketch Dimension** tool.
3. Create the main reference lines and locate the required points according to the dimensions given in the question. A horizontal distance of **47 mm** is maintained between the required centre positions, and the vertical dimensions are applied as specified in the drawing.
4. Use the **Circle** tool to create the required circular features. Draw the circles at the specified locations and apply their respective diameters, including **Ø20, Ø32, Ø10 and Ø20**, as shown in the given drawing.
5. Construct the upper circular portion of the component by creating the required concentric circles. Apply the corresponding dimensions and use the centre points to maintain the required alignment.



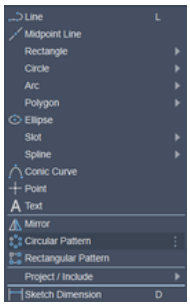
6. Next, create the outer profile of the component using the required **Line** and **Arc** tools. The curved portions are formed according to the dimensions given in the drawing, including the required **R64** and **R20** radii.
7. Apply the required geometric constraints and dimensions to position the curves correctly with respect to the previously created circles and reference geometry. This ensures that the sketch follows the proportions and measurements of the given component.

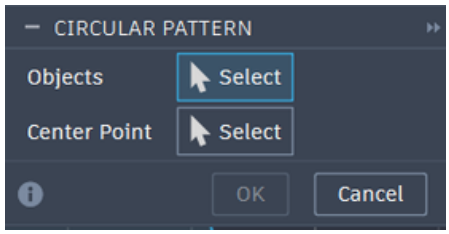


8. Create the two side circular features using the**Circle** tool. Position them according to the given horizontal and vertical dimensions andapply the required radius values, including **R7 and R10**.
9. Create the lower circular feature at the centreofthe bottom portion. Draw the required circles and apply the specified **Ø20** and **Ø10**dimensions.
10. Construct the curved connections betweenthelower circular features and the main outer profile using the **Arc** tool. Apply the requiredradius values and adjust the curves so that they smoothly connect with the surroundinggeometry.
11. After creating the basic geometry, use the **Mirror**option to reproduce the required symmetrical geometry about the appropriatecentre/reference line. Select the geometry that has to be mirrored and choose the corresponding centre line as the mirror axis.

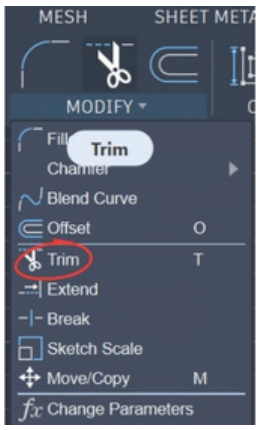


12. Once the mirrored geometry is created, check both sides of the sketch to ensure that the corresponding features are symmetrical and correctly positioned.
13. Create the central patterned feature using the required circles and arcs. Draw the central circle and construct the surrounding repeated curved portions according to the dimensions shown in the drawing.
14. Apply the required dimensions to the central feature, including **Ø40, Ø66 and R6**, and use the appropriate pattern/mirror operation to obtain the repeated geometry.



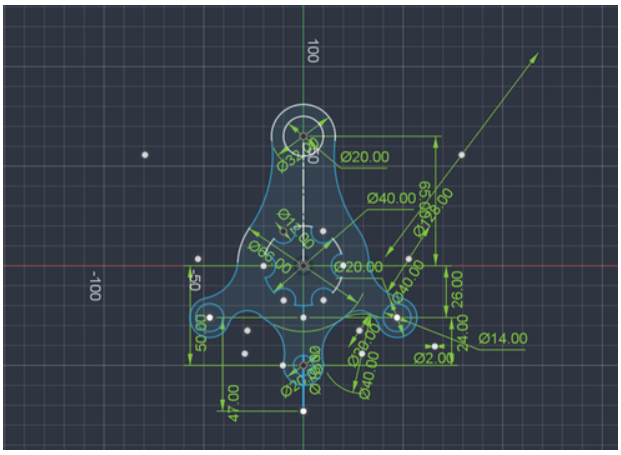


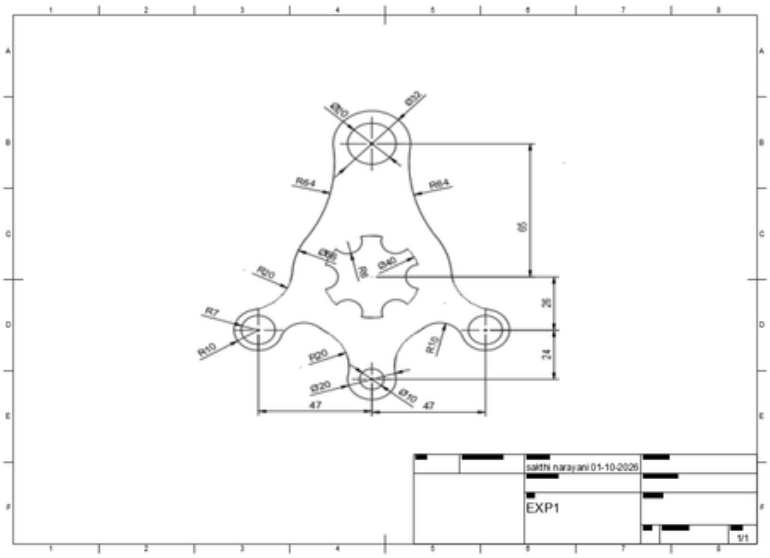
15. After completing the required geometry, use the **Trim** tool to remove the unwanted portions of circles, arcs and overlapping lines. Select the portions that are not part of the final profile and trim them carefully.



16. Continue trimming the unnecessary sections until the outer boundary and inner features match the given component drawing. Check that no unwanted construction or overlapping portions remain in the final sketch.
17. Verify all important dimensions and positions in the completed sketch. Check the **65 mm** vertical dimension, **47 mm** horizontal dimensions, the **26 mm and 24 mm** vertical dimensions, and the specified radii and diameters.
18. Make the necessary corrections wherever required and ensure that the final geometry matches the reference drawing. The completed sketch should contain the upper circular portion, side holes, lower circular feature, outer curved profile and central patterned feature.
19. Finally, zoom out and display the **complete drawing sheet** containing the finished 2D sketch and its dimensions. Take a screenshot of the complete drawing for documentation.
20. Save the completed Fusion design. The final sketch is then ready to be included in the experiment documentation.

OUTPUT:





RESULT:

Thus the given sketch is drawn and drafted using Fusion 360 tool.